SAMANVITHA ADARASUPALLY

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PROFESSIONAL SUMMARY

Computer Engineering student (CGPA 8.52/10) with hands-on experience in Python, DSA, DBMS, and machine learning through academic projects and two AI/Data Science internships. Skilled in building CNN-based computer vision systems, IoT solutions, and REST-based web applications. Seeking a Specialist Programmer / Digital Specialist Engineer (Trainee) role to apply strong CS fundamentals and problem-solving ability.

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, SQL
- **Core CS Fundamentals:** Data Structures & Algorithms, OOPs, Operating Systems, Computer Networks, DBMS
- **Web & Systems:** REST APIs, Microservices (fundamentals), Git & SDLC, HTML/CSS/JavaScript
- **Databases & Tools:** MySQL, NoSQL concepts, MATLAB, Google Colab, Jupyter Notebook, MS Excel/Word/PowerPoint
- **Soft Skills:** Critical Thinking, Root Cause Analysis, Communication, Team Collaboration, Adaptability

WORK EXPERIENCE

AI Research Intern — Symbiosis University *Ongoing*

- Contributing to applied AI research, model development, and experimentation with emerging AI techniques.

AI & Data Science Intern — ASD Solutions *Completed*

- Performed data analysis and built machine learning models to support AI-driven project deliverables.

PROJECTS

Bi-Directional Sign Language Communication System | Python, OpenCV, CNN, MATLAB *github.com/Samanvitha-A*

- Built a real-time sign language recognition system using CNN and computer vision, achieving ~90% accuracy.
- Integrated hand tracking, voice-to-text conversion, and a GUI for two-way sign-to-speech communication; presented at Hack India Hackathon.

Face Mask Detection AI | Python, OpenCV, CNN

- Developed a computer vision model to detect face mask usage in real time from images, video, and live camera feeds.

VPN Tunneling System | Python, Networking

- Implemented an encrypted VPN tunnel to securely route data between two endpoints over a public network.

Dynamic Ambulance Traffic Signal Overriding System | IoT, LoRa

- Designed an IoT-based system to auto-prioritize ambulances at traffic signals with multi-junction coordination and real-time hospital alerts, reducing emergency response time.

EDUCATION

Chaitanya Bharathi Institute of Engineering and Technology — B.E. Computer Engineering *Aug 2023 – Present*

- CGPA: 8.52 / 10

Narayana Junior College — M.P.C (Mathematics, Physics, Chemistry) *Jun 2021 – Mar 2023*

- Score: 940 / 1000 (94%)

CERTIFICATIONS

- Python Programming
- Machine Learning — NPTEL, IIT Madras
- Industry 4.0: Industrial Internet of Things (IoT) — NPTEL, IIT Kharagpur

ACHIEVEMENTS & LEADERSHIP

- Hack India Hackathon: Built a bi-directional sign language communication system for real-time sign-to-text/speech conversion under tight deadlines.
- Volunteering: Active weekend member for street cause events, organizing public fundraising activities for the underprivileged.